

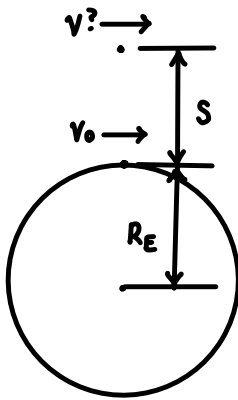
AERO40007 – Mechanics

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Orbital Mechanics

(1.) An object is projected vertically upwards from the North pole with initial speed v_0 . The radius of the Earth is R_E . Neglecting air resistance, find the speed at a distance s above the Earth's surface.

Why are spacecraft usually launched from locations near to the equator?



$$E_{v_0} = E_v = E_k + E_p = \frac{1}{2}mv^2 + mgh$$

$$\therefore \frac{1}{2}mv_0^2 - \frac{GMm}{R_E} = \frac{1}{2}mv^2 - \frac{GMm}{R_E + s}$$

$$\therefore v^2 = v_0^2 + \frac{2GM}{R_E + s} - \frac{2GM}{R_E}$$

$$\therefore v^2 = v_0^2 + \frac{2GMS}{R_E(R_E + s)}$$

$$\therefore v = \sqrt{v_0^2 + \frac{2GMS}{R_E(R_E + s)}}$$

$$mg = \frac{GMm}{R_E^2}$$

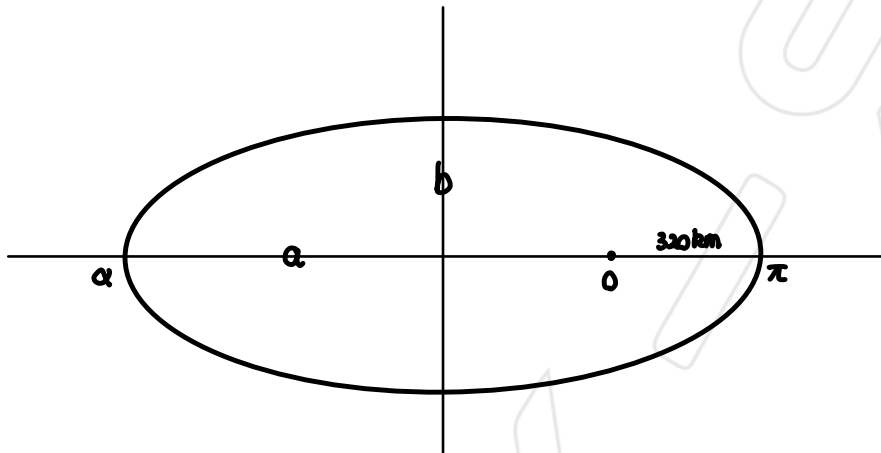
$$\therefore g = \frac{GM}{R_E^2} \quad \therefore GM = gR_E^2$$

$$\therefore v = \sqrt{v_0^2 + \frac{2gR_E s}{R_E + s}}$$

Extra v_0 will be provided at equator due to the Earth's rotation

(2.) A satellite is in an orbit with eccentricity $e = 0.5$ around Mars. At the periapsis it is 320 km above the surface of the planet. The radius of Mars $R_M = 3390$ km (just over half the radius of Earth) and the acceleration due to gravity on the surface is $g_M = 3.78 \text{ m/s}^2$. Find:

- The maximum velocity of the satellite relative to the centre of Mars.
- The period of the orbit.
- The maximum distance of the satellite from the surface of Mars.
- The escape velocity at the periapsis.



(i) $V_{\max}$ at π ,

$$\therefore V_{\max} = V_{\pi} = \frac{h}{r_{\pi}} = \frac{h}{\frac{h^2}{GM(1+e)}} = \frac{GM(1+e)}{h}$$

$$\therefore g = \frac{GM}{r^2} \quad \therefore V_{\max} = \frac{gR_M^2(1+e)}{h} = \frac{gR_M^2(1+e)}{r_{\pi} V_{\max}}$$

$$\therefore V_{\max}^2 = \frac{gR_M^2(1+e)}{r_{\pi}} = 4190.87 \text{ m/s}$$

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- (ii) The period of the orbit.
- (iii) The maximum distance of the satellite from the surface of Mars.
- (iv) The escape velocity at the periapsis.

$$\text{ii) } GM = g_M R_M^2 = \left(\frac{2\pi}{T}\right)^2 a^3 = \left(\frac{2\pi}{T}\right)^2 \left(\frac{1}{2}(r_p + r_a)\right)^3 = \frac{\pi^2 (r_p + r_a)^3}{2T}$$

$$\therefore T = \sqrt{\frac{\pi^2 (r_p + r_a)^3}{2g_M R_M^2}}$$

$$\therefore r_a = \frac{h^2}{GM(1-e)}$$

$$r_p = \frac{h^2}{GM(1+e)}$$

$$\therefore r_a = \frac{r_p(1+e)}{(1-e)}$$

$$\therefore T = 19268.15 \text{ s}$$

$$\text{iii) } r_a = 11130 \text{ km}$$

$$r_{\text{max}} = r_a - R_M = 7740 \text{ km}$$

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iv) escape $e = 1$

$$\begin{aligned} \therefore v_{\lambda} \text{ (when } e=1) &= \sqrt{\frac{gR_M^2(1+e)}{r_{\lambda}}} = \sqrt{\frac{2gR_M^2}{r_{\lambda}}} \\ &= 4839.20 \text{ m/s} \end{aligned}$$

(3.) A meteor is moving at a speed of 32000 km/hr relative to the centre of Earth at a point 560 km from the surface of the Earth. At this time it has a radial velocity of 6400 km/hr. How close does it come to the Earth's surface? ($R_E = 6373$ km)

$$E = \frac{1}{2}mv^2 + mgh = \frac{1}{2}mv^2 - \frac{GMm}{r} = \text{constant}$$

$$v_\phi^2 = v^2 - v_r^2, \quad d = 560 \text{ km}$$

$$h = (R_E + d) \cdot v_\phi = r_\pi v_\pi = \text{constant}$$

$\therefore \pi$ is closest to the surface

$$\therefore v_\pi^2 - \frac{2GM}{r_\pi} = v^2 - \frac{2GM}{R_E + d}$$

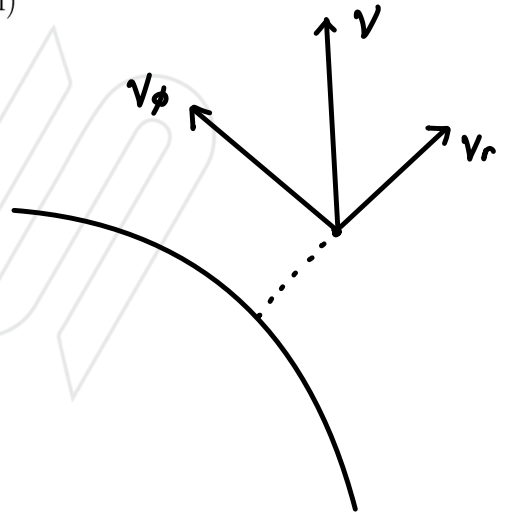
$$\therefore \frac{2gR_E^2}{r_\pi} = -v^2 + v_\pi^2 + \frac{2gR_E^2}{R_E + d}$$

$$\therefore v_\pi = \frac{h}{r_\pi} \quad \therefore \frac{2gR_E^2}{r_\pi} = \frac{h^2}{r_\pi^2} - v^2 + \frac{2gR_E^2}{R_E + d}$$

$$\therefore \left(\frac{2gR_E^2}{h} - v^2 \right) r_\pi^2 - (2gR_E^2)r_\pi + h^2 = 0$$

$$\therefore r_\pi =$$

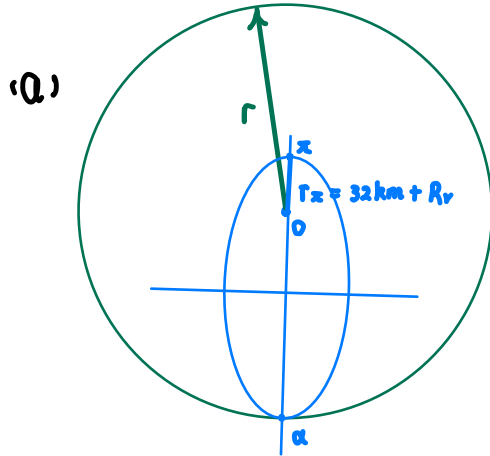
$$\therefore d_{\min} =$$



(4.) A spacecraft is in a circular parking orbit of radius 320 km around the planet Venus. The radius of Venus $R_V = 6196$ km, and the escape velocity at the surface is 1.026×10^4 m/s. A retrorocket is fired to slow the spacecraft so it will come to within 32 km of the surface.

(a) If the rocket changes the speed of the spacecraft over a short time compared to its orbital period, what is the change in velocity required?

(b) What is the speed of the spacecraft, relative to the centre of Venus, at the periapsis of its new orbit?



$$v_{esc} = \sqrt{\frac{2GM}{r}} = 1.026 \times 10^4 \text{ m/s}$$

$$\therefore GM = \frac{1}{2} (v_{esc})^2 \cdot (6196 \text{ km} + 320 \text{ km})$$

$$\therefore e = 0$$

$$\therefore r = \frac{h_i^2}{GM(1+0)} = \frac{h_i^2}{GM}$$

$$\therefore h_i = v_o r$$

$$\therefore r = \frac{v_o^2 r^2}{GM} \quad \therefore v_o = \sqrt{\frac{GM}{r}} = 7254.91 \text{ km/s}$$

$$r_a = r$$

$$\therefore \frac{1}{2} v_x^2 - \frac{GM}{r_x} = \frac{1}{2} v_a^2 - \frac{GM}{r_a}$$

$$\therefore h_x = v_x r_x = v_a r_a = v_o r$$

$$\therefore v_a =$$

$$\therefore \Delta V = v_a - v_o =$$

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b. v_π

Answers

1.

$$v = \sqrt{v_0^2 - \frac{2gR_E s}{R_E + s}}$$

2.

(i) 15080 km/hr, (ii) 5.35 hrs, (iii) 7728 km, 17420 km/hr

3.

$$r_{min} = 6453 \text{ km}$$

4.

$$\Delta v = 290 \text{ km/hr}, v_\pi = 26340 \text{ km/hr}$$